

Deepak Mallampati

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Kent, OH (US-based, F-1 OPT, open to sponsorship)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) who owns the full AI/ML solution-delivery cycle - translating business needs into Python-backed LLM applications and predictive ML pipelines, packaging with FastAPI + Docker, monitoring in production, and translating technical limitations into stakeholder-facing system-limitation docs. Healthcare RAG with ~35% retrieval precision, agentic LLM workflows with >30% hallucination reduction, predictive ML (scikit-learn / XGBoost) with 15-20% recall gains on high-risk categories, and ~30% post-deployment defect reduction via structured logging and load simulation. Comfortable across data, models, APIs, and Docker-based deployment in HIPAA-conscious environments.

SKILLS

AI/ML Solution Delivery • LLM Applications & Agentic Workflows • Predictive ML (scikit-learn, XGBoost) • RAG & Grounding • FastAPI / Docker Production • Cloud-Ready Deployment • CI/CD & Model Monitoring • Stakeholder Communication

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

LLM & GenAI: LLM application development, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search, responsible AI

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion, video analytics

Data & Analytics: Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation, ETL pipelines, NetworkX

Infra & DevOps: Docker, Jenkins CI/CD, Grafana, MLOps/LLMOps, RESTful APIs, cloud-ready deployment, structured logging / observability

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Multimodal Systems

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - USA (Healthcare Technology, AI Consulting)

- Translated clinical-ops business needs into **Python LLM applications and agentic workflows** (eligibility checks, care workflow navigation, documentation clarification); agent response stability +~25%, hallucination reduction >30%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement, support prioritization; recall +15-20% on high-risk categories at precision >90% via class weighting, stratified sampling, threshold calibration.
- Built **RAG systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings; contextual retrieval precision +~35%, grounded-response alignment >90%.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, structured logging + load simulation; post-deployment defects ~30%.
- Built **ETL preprocessing pipelines** (Pandas, NumPy) for EHR extracts, appointment histories, support ticket logs; dataset reliability >98%, downstream model instability ~40%.
- Wrote stakeholder-facing system-limitation docs alongside every release - the artifact that earned trust faster than improving the model itself.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India

- Built **CI/CD pipelines with Jenkins** for schema updates, version-controlled SQL scripts, stored procedure deployments; deployment errors **->30%**, release cycle time **-35-40%**.
- Designed performance dashboards using **SQL Server DMVs and Grafana** for long-running queries, deadlocks, CPU/I/O contention; incident recurrence **~25%**, root-cause resolution **+~30%**.
- Optimized SQL and **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform; query execution time **~20%**, data retrieval latency **~25%**.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Hyderabad, India (Video Analytics, Applied NLP)

- Built a **transformer-based video summarization and highlight-generation system**; manual review time **-60-70%**, ~85% highlight alignment with human curation.
- Implemented **document Q&A with RAG**; hallucinated outputs **~30%**, contextual answer accuracy **>85%**.
- Deployed as a **Flask-based API** with a lightweight web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake -> Validation -> Consistency -> Duplicate -> Fraud Risk Scoring) with **schema-validated JSON contracts between agents**. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity search for duplicate detection; explainable risk scoring with audit-ready reasoning traces - full SE-style delivery archetype.

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation (shipped, Chrome Web Store)

Full-stack browser extension built independently - Manifest V3, content scripts, service workers. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Per-provider payload handling; 7-day cache; 29 site configs; privacy policy; narrowed host permissions.

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating multimodal APIs (Perplexity Sonar + Replicate).

Intermediate structured prompt transformation layers improved contextual alignment **~30%** and first-pass image success rate **~25-30%**.

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced a two-stage CNN pipeline with a **unified YOLOv8** model; inference latency **~30%**, false positives **~25%** via sliding-window confidence aggregation; adaptive frame skipping; NMS tuning.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025